

Dexter Fernandes

SLAM Computer Vision Engineer

Fluent in English

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PROFESSIONAL SUMMARY

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for **Tier 1 companies**, including a construction firm in Texas and a UK oil and gas company. Currently pursuing an MSc in Robotics at the University of Bristol, specialising in **LiDAR SLAM and odometry** for a Mobile Radiological Survey Robot. Proven track record of collaborating with **multinational clients** to deploy AI-driven vision solutions in **fast-paced start-up** environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes. Passionate about advancing **localisation and mapping** for autonomous systems, and actively seeking opportunities to further specialise in SLAM technologies.

SKILLS

Deep Learning | Computer Vision | Time-series Analysis | Machine Learning | Test Driven development(TDD) | Version Control | Containerisation | MLOps | Distributed Event Data-Streaming | Project Management | Simultaneous Localisation and Mapping | LiDAR | API Development | Troubleshooting | Leadership and Management | Client Interaction

DISSERTATION

University of Bristol

MSc Robotics - Dissertation

Bristol, England

June 2024 – September 2024

Radiological SLAM

- **Led the full development** of a UGV system on the Clearpath Husky, integrating LiDAR SLAM for real-time mapping and radionuclide localisation to build dense 3D point cloud maps, delivering an end-to-end solution.
- Integrated Velodyne LiDAR, Symetrica Verifinder, and Intel Realsense with complex network communications, enabling **radiological mapping** and radionuclide detection.
- **Developed low-latency APIs** to extract real-time data from the Verifinder, ensuring accurate measurements during testing, while containerising the system in Docker for efficiency.
- Achieved precise radionuclide localization using **advanced odometry techniques** like Scan-to-Map ICP, significantly reducing drift and enhancing detection accuracy in real-time LiDAR SLAM operations.
- Led software development and API integration using **ROS2** and **Docker**, improving system modularity and scalability for real-time LiDAR SLAM deployment on the Nvidia Jetson AGX Xavier.

WORK EXPERIENCE

Esbaar

AI Engineer

Muscat, Oman

January 2023 – August 2023

Z-Alert

- **Developed custom time-series models** to classify 15 distinct oil rig operations while managing and analysing 3.6 million sensor data points covering one well cycle.
- Optimised hyperparameter tuning for time-series models by **integrating MLOps**, orchestrating the training of 500 models, and boosting the F1-score by 39%.
- **Led the development of vision models** using YOLOv5 to detect rig workers and heavy equipment, improving F1-score by 17% over the previous solution through model iteration, maintenance, and mitigating model degradation.
- Leveraged **Apache Kafka with Docker containers** to re-factor a monolithic application into a **microservices** architecture, reducing software update intervals and improving customer satisfaction.
- Key achievement: **Resolved a year-long segmentation fault** in the legacy monolithic application within one day, demonstrating exceptional problem-solving skills and technical expertise.

LivNSense Technologies Pvt. Ltd.

Lead Software Engineer - AI

Bengaluru, India

May 2020 – December 2022

ViCAS – Safety-based Anti-Collision Early Detection System

- **Managed a team of 2-4 persons** that contributed to the entire pipeline – curating the training and testing datasets, managing model life cycle in production, and tackling solution edge cases.
- **Boosted vision model inference speed** by 3.6x on the same hardware, using advanced optimisation techniques such as mixed-precision inference and deep learning expertise to dramatically improve performance and efficiency.
- Surpassed the existing vision solution by revamping the dataset, re-formulating the problem statement, and **reducing inference time** from 250 ms to 90 ms on the same edge hardware.
- **Achieved faster model deployment and significantly reduced operational costs** by reducing model training time from 2.5 days to 4 hours on the same hardware, demonstrating expertise in deep learning frameworks and advanced optimisation techniques.
- **Overhauled the labelling platform** and repurposed older production models for **auto-labelling**, increasing annotator output from 600 to 3000 images per day and maximising resource efficiency.
- Represented the company as a consultant abroad, **understanding client requirements** and delivering a **pilot solution** with real-time analytics within two weeks. – Workplace Hazard Detection System

EDUCATION

University of Bristol

MSc in Robotics

Bristol, England

September 2023 – September 2024

Visvesvaraya National Institute of Technology

B. Tech in Electrical and Electronics Engineering

Nagpur, India

July 2016 – May 2020

SOFTWARE STACK

Python | C++ | OpenCV | PyTorch | TensorFlow | ROS | Intel OpenVINO | Sci-kit Learn | MLFlow | Optuna | Github | Docker | Jira | CVAT | Matplotlib | Shell Scripting (Linux) | Git

CERTIFICATIONS

Deep Learning Specialisation – 5-course specialisation at deeplearning.ai

AWARDS

Received the **Best Speaker award** on 3 occasions from ABC Toastmasters club in Muscat, Oman.